Audrey Horn

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EDUCATION

Cornell University **Expected 2029**

Major: Applied & Engineering Physics

- **Relevant Coursework:** Mechanics & Special Relativity; Multivariable Calculus; Differential Equations; Electricity & Magnetism; Experimental Physics; Lasers & Photonics; Quantum Information Hardware; Computer Science; Linear Algebra; Waves & Oscillations; Digital Logic & Computer Organization.

TECHNICAL SKILLS

Programming & Data Analysis: Python (NumPy, SciPy, Pandas, Matplotlib, Astropy, Qiskit), Java, Embedded C/C++, Git/GitHub, Linux, LaTeX.

Digital & Embedded Systems: Verilog, Quartus Prime, FPGA Design, Synthesis, Simulation & Verification, Digital Logic, CMOS Transistor-Level Logic, ESP32 (Arduino Framework), Stepper Motors, Motor Drivers.

Hardware & Experimental Techniques: Pulsed Laser & Optical Systems, Oscilloscopes, Circuit Prototyping & Troubleshooting, Schematic Interpretation, Soldering, Electromechanical Integration.

CAD, Simulation & Fabrication: SolidWorks, Ansys (FEA), Lathe, Mill, Bandsaw (Certified User).

Design & Visualization: Blender, Adobe Photoshop, Adobe Premiere Pro, Adobe Lightroom, Adobe Illustrator.

RESEARCH & TECHNICAL EXPERIENCE

Cornell University, Quantum Computing Association **Ithaca, New York**

Theory/Decoding Subteam Member *Apr 2026 – Present*

- Study quantum error-correction decoders (belief propagation, MWPM/Blossom, Union-Find), including their complexity and suitability for fault-tolerant computing.
- Study qubit error models (bit-flip, phase-flip, decoherence) and their effect on syndrome measurement and decoding accuracy.
- Work with the hardware subteam on FPGA-based decoder implementations, evaluating latency, throughput, logic utilization, and real-time constraints on syndrome processing and feedback.
- Study superconducting qubit physics: transmon architecture, Josephson junction nonlinearity, energy-level structure, and relaxation and dephasing.
- Model single-qubit dynamics analytically and in simulation, including Rabi oscillations, Ramsey interferometry, rotating-frame dynamics, detuning, and dynamical decoupling (CPMG, XY4).

Cornell Hyperloop Project Team **Ithaca, New York**

Magnetics Hardware Engineer *Nov 2025 – Present*

- Designed, built, and debugged an ESP32-controlled coil-winding system using NEMA 17 steppers, DRV8825 drivers, and embedded C/C++ for repeatable electromagnet fabrication.
- Implemented configurable winding speed, turn count, direction, and wire traverse; debugged motor synchronization, wiring, and mechanical alignment.
- Designed and built electromagnet assemblies (ferromagnetic cores, wound coils, custom housings) for the pod's electromagnetic suspension system.
- Calculated coil geometry, winding configuration, current, magnetic flux, and air gap for the suspension electromagnets.
- Fabricated and assembled electromagnetic components for the linear induction motor and helped prepare it for testing.
- Designed electromechanical parts in SolidWorks and evaluated them with Ansys FEA for structural load, materials, tolerances, and manufacturability.
- Worked across magnetics and mechanical subteams on pod integration; team placed 5th in the Design-only division at European Hyperloop Week 2026.

Cornell University, Lasers & Photonics, AEP 1100 **Ithaca, New York**

Laboratory Assistant *Aug 2026 – Present*

- Support students building and operating pulsed nitrogen lasers, including the pulse generator, pulse amplifier, and laser chassis.
- Help students assemble and troubleshoot circuits and laser hardware.
- Guide students in measuring laser pulses with oscilloscopes and photodiodes and in analyzing results and experimental uncertainty.
- Enforce high-voltage and laser safety, supervise lab sessions, and give feedback on calculations, documentation, and assignments.

Cornell Extreme Astrophysics Group **Ithaca, New York**

Undergraduate Researcher (Nexus Scholar, Summer 2026) *May 2026 – Present*

- Evaluated 200+ FRB source datasets and curated a validated catalogue of 126 events with observational parameters, source references, and measurement uncertainties.
- Built Python pipelines for data ingestion, normalization, coordinate conversion, uncertainty propagation, and source traceability.
- Recomputed extragalactic dispersion measures using the NE2025 Galactic electron-density model to remove the Milky Way contribution.
- Produced scattering-DM and DM-redshift diagnostics to compare observed FRB properties with theoretical propagation predictions.
- Reproduced the host-galaxy analysis of Cordes, Ocker & Chatterjee (2022) in Python, combining the Macquart relation, IGM cosmic variance, and a cloudlet scattering model to constrain host DM and scattering time vs. host redshift for individual catalogue FRBs.

SELECTED ENGINEERING PROJECTS

5-Bit FPGA Numeric Display | Digital Logic and Computer Organization | *Verilog, Quartus Prime, FPGA*

- Implemented a gate-level Verilog display unit converting a 5-bit binary input (0–31) to two decimal digits on seven-segment displays, using binary-to-BCD and seven-segment decoders.
- Built unoptimized (sum-of-products) and Karnaugh-map-optimized decoders and compared their area and minimum critical-path delay after synthesis.
- Built an add/multiply calculator on top of the display (2-bit × 16-bit multiplication), hierarchically from full adders, ripple-carry and carry-select adders, 2:1 muxes, and a partial-product multiplier.
- Prototyped both designs on an Intel Cyclone V FPGA (DE0-CV) in Quartus Prime, with six seven-segment displays and a GPIO-driven overflow LED.
- Compared ripple-carry, carry-select, and RTL adders by tightening timing constraints in 1 ns steps to find each design's minimum critical-path delay, and traced the critical paths through individual gates.
- Verified designs with exhaustive, directed, random, and X-propagation testbenches (Icarus Verilog, Verilator), debugging with Surfer waveforms.
- Breadboarded a full adder and a 2-bit ripple-carry adder from 74-series ICs and measured gate delays on an oscilloscope.

Nitrogen Laser Pulse Amplifier | Lasers and Photonics | *High-Voltage Electronics, SCRs, Oscilloscope*

- Built and tested the pulse amplifier for a 337.1 nm nitrogen laser, the middle of three cascaded subsystems (pulse generator → pulse amplifier → thyratron discharge chassis).
- The amplifier converts an 18 V, ~3 Hz 555-timer trigger into a ~1.6 kV fast-rise pulse that fires an EG&G HY-32 hydrogen thyratron, discharging a capacitor bank through the laser channel at 10–25 kV.
- Built the ~400 Vdc supply: a center-tapped full-wave rectifier off a 270 Vac secondary, with 40 μF and 0.01 μF / 1 kV filtering and a 1 MΩ bleeder.
- Wired the switching stage, where a 20 kΩ, 5 W resistor charges a 1.0 μF capacitor that an NTE5416 SCR discharges through a 1:4 pulse transformer, stepping ~400 V up to ~1.6 kV.
- Implemented the SCR turn-off network (0.47 μF capacitor, 1N914 diode string) and a ~100:1 divider for safe oscilloscope monitoring.
- Soldered the circuit on flea-clip boards separated by voltage domain, with single-point grounding and service-length wiring for repair access.
- Verified pulse amplitude and timing on the oscilloscope before integration; the finished laser was characterized for lasing threshold, ~10 ns pulse width, and output vs. N_2 pressure (5–65 torr).